

HR Analytics Dashboard

Power BI Project Report

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1. Executive Summary

This report documents the development and findings of an HR Analytics Dashboard built in Microsoft Power BI. The project uses a dataset of 1,000 employee records to explore workforce composition, salary structure, performance, attrition trends, and hiring history. The primary objective was to transform raw HR data into a coherent set of interactive visualizations that allow HR professionals and business leaders to monitor workforce health and make informed, data-driven decisions.

The dashboard was designed around ten core business questions — covering department size, attrition rates, gender balance, salary levels, age distribution, and the relationship between performance and compensation. Using Power Query for data preparation and DAX for calculated measures, the report delivers a comprehensive view of the organization's human capital.

Key Findings at a Glance

Metric	Value
Total Employees	1,000
Average Salary	\$55,369
Overall Attrition Rate	50.30%
Average Experience (Years)	11.46
High Performance Employees	380
Largest Department	Marketing (211 employees, 21.1%)
Highest Attrition Rate by Dept.	Finance (54.00%)
Gender Split	Female 50.9% / Male 49.1%
Most Common Age Group	Older (45+), with 462 employees

The dashboard proves particularly useful for HR decision-making because it consolidates multiple workforce dimensions into a single, filterable interface. Slicers for department, gender, age group, performance score, and salary level allow users to drill into any segment of the workforce and quickly identify patterns that may otherwise remain hidden in spreadsheets.

2. Introduction

Human Resources analytics has grown from a back-office function into a strategic discipline. Organizations that can measure and interpret workforce data are better positioned to reduce unnecessary turnover, align compensation with performance, and plan for future hiring needs. Despite this, many HR departments still rely on static spreadsheets that offer limited insight into trends or cross-functional comparisons.

This project was developed in response to that gap. By building an interactive Power BI dashboard, the goal was to enable HR stakeholders to explore employee data dynamically — filtering by department, salary level, gender, or age group — without requiring technical expertise. The dashboard was designed to answer practical HR questions rather than simply display data.

Attrition, in particular, is one of the most costly challenges an organization can face. Losing experienced employees means losing institutional knowledge, and the cost of recruitment, onboarding, and training can be substantial. Understanding which departments, salary groups, or experience bands are most affected by attrition is therefore a business-critical concern.

Similarly, salary analysis and performance tracking are essential tools for maintaining employee satisfaction and motivation. If high-performing employees are not compensated accordingly, the risk of voluntary departure increases. The dashboard examines this relationship directly and highlights areas where HR may want to investigate further.

In summary, the goal of this project was to produce a professional, interactive HR dashboard that supports evidence-based decision-making across five key dimensions: employee distribution, salary structure, performance, attrition, and hiring trends.

3. Dataset Description

The dataset used in this project is a CSV file named `employees_data.csv`, containing records for 1,000 employees. Each row represents one employee and includes the following fields:

Column	Data Type	Description
EmployeeID	Integer	A unique numeric identifier for each employee. Used to remove duplicates during data preparation.
Name	Text	The employee's full name. Not used analytically, but included in the Top 10 by Salary table.
Gender	Text (Male/Female)	The employee's gender. Used to analyze gender distribution and gender-based attrition.
Age	Integer	The employee's current age in years. Used to group employees into age categories.
Department	Text	The business department (Marketing, HR, Finance, Sales, IT). Central to most comparisons.
JobTitle	Text	The employee's specific role. Used in the Average Salary by Job Title visualization.
HireDate	Date	The date the employee joined the organization. Used to calculate years of experience.
Salary	Integer (USD)	Annual salary in US dollars. Used in salary distribution, averages, and salary level grouping.
PerformanceScore	Integer (1–5)	A numeric score reflecting employee performance. Used in the scatter plot and KPI cards.
Attrition	Text (Yes/No)	Indicates whether the employee has left the organization. The key outcome variable throughout the analysis.

The dataset provides enough breadth to explore cross-dimensional relationships — for example, whether attrition is more common in lower salary brackets, or whether older employees have significantly more experience variation than younger ones. The combination of categorical variables (Department, Gender, AgeGroup) and numeric variables (Salary, PerformanceScore, ExperienceYears) makes it well-suited for both descriptive and comparative analysis.

4. Data Preparation

All data cleaning and transformation was completed within Power Query, the built-in data preparation environment in Power BI. The following steps were taken before any analysis or visualization was performed.

4.1 Removing Duplicates

The first step was to remove any duplicate records based on the EmployeeID column. Since each EmployeeID should correspond to exactly one employee, any repeated values would distort aggregate counts and measures. Power Query's 'Remove Duplicates' function was applied to the EmployeeID column to ensure data integrity from the outset.

4.2 Converting HireDate to Date Format

The HireDate column was originally imported as plain text in some cases. It was converted to a proper Date data type to enable date-based calculations. This step was necessary for Power BI to correctly interpret the column and support derived columns such as ExperienceYears.

4.3 Creating the ExperienceYears Column

A calculated column named ExperienceYears was created using the following logic: the number of full years elapsed between the HireDate and the current date (today's date at the time of data refresh). This was implemented in Power Query using:

```
ExperienceYears = Date.Year(DateTime.LocalNow()) -  
Date.Year([HireDate])
```

This column was essential for analyzing how tenure relates to attrition and for the Experience Distribution by Age Group boxplot visualization.

4.4 Creating the SalaryLevel Column

The SalaryLevel column was created as a categorical grouping of the numeric Salary field. This allows for simplified comparison across compensation tiers and enables the Attrition by SalaryLevel analysis. The classification rules are:

SalaryLevel	Salary Range
Low	Below \$40,000
Medium	\$40,000 – \$70,000
High	Above \$70,000

Without this grouping, salary data would only be analyzable as a continuous variable, which is less intuitive for HR stakeholders who think in terms of compensation bands.

4.5 Creating the AgeGroup Column

Similarly, the AgeGroup column was derived from the Age field to group employees into three broadly meaningful demographic categories:

AgeGroup		Age Range	
Young		18 – 29 years	
Middle Age		30 – 44 years	
Older		45 years and above	

These groupings make it easier to compare workforce composition and attrition patterns across life stages, and they serve as both a slicer and a dimension in multiple visualizations.

5. DAX Measures

DAX (Data Analysis Expressions) is the formula language used in Power BI to create calculated measures. Each measure below was written to support a specific KPI or visualization in the dashboard. The following table summarizes all measures created in this project:

5.1 Total Employees

This measure counts all employee records in the dataset. It is the foundation for virtually every other measure in the dashboard, including attrition rate and high performance percentage.

```
Total Employees = COUNT(employees_data[EmployeeID])
```

5.2 Average Salary

Calculates the mean salary across all employees currently visible in the filter context. This measure responds dynamically to slicer selections.

```
Average Salary = AVERAGE(employees_data[Salary])
```

5.3 Total Attrition

Counts only those employees where the Attrition field is marked as 'Yes'. This is the numerator in the attrition rate calculation.

```
Total Attrition =  
CALCULATE(  
    COUNT(employees_data[EmployeeID]),  
    employees_data[Attrition] = "Yes"  
)
```

5.4 Attrition Rate

Divides Total Attrition by Total Employees to produce a percentage. The DIVIDE function is used instead of a simple division operator to safely handle cases where the denominator might be zero.

```
Attrition Rate = DIVIDE([Total Attrition], [Total Employees])
```

5.5 Average Experience Years

Returns the mean value of the ExperienceYears column, giving a quick summary of tenure across the workforce.

```
Avg Experience Years = AVERAGE(employees_data[ExperienceYears])
```

5.6 High Performance Employees

Counts employees with a PerformanceScore of 4 or higher. This measure helps HR quickly assess how much of the workforce is performing at a high level.

```
High Performance Employees =  
CALCULATE(  
    COUNT(employees_data[EmployeeID]),  
    employees_data[PerformanceScore] >= 4  
)
```

5.7 Female Percentage

Calculates the proportion of female employees in the current filter context. Used to present gender balance as a percentage.

```
Female % =  
DIVIDE(  
    CALCULATE(COUNT(employees_data[EmployeeID]),  
        employees_data[Gender] = "Female"),  
    [Total Employees]  
)
```

All measures are filter-context aware, meaning they respond automatically to any slicer selection the user makes on the dashboard. This ensures that every KPI and chart updates consistently whenever a filter is applied.

6. Dashboard Design and Visualizations

The dashboard was structured across three visual panels, each serving a distinct purpose. Rather than cramming all charts onto a single canvas, the layout was organized so that general summary metrics appear at the top, employee distribution in the middle, and deeper analytical comparisons at the bottom. This hierarchy guides the user from a high-level overview toward more specific insights.

6.1 Header and Slicers

The header bar at the top of the dashboard displays the title 'HR Analytics Dashboard' and contains five dropdown slicers: Department, Gender, AgeGroup, PerformanceScore, and SalaryLevel. These slicers are cross-filtering, meaning that any selection instantly updates all KPI cards and charts on the page. This makes the dashboard fully interactive and suitable for ad-hoc exploration without any technical knowledge.

6.2 KPI Cards

Directly below the header, five KPI cards display the most important workforce metrics at a glance. Each card shows a single number: Total Employees (1K), Average Salary (\$55,369), Attrition Rate (50.30%), Average Experience Years (11.46), and High Performance Employees (380). These cards serve as the dashboard's 'vital signs' — the first metrics any HR professional or executive would want to see.

6.3 Employee Hiring by Year (Line Chart)

This line chart shows how many employees were hired in each calendar year, from approximately 2005 to 2024. It provides a historical view of recruitment activity and can reveal whether the organization has been growing steadily, experienced hiring freezes, or ramped up recruitment in specific periods. The chart shows noticeable fluctuation year over year, which may reflect changing business conditions or strategic headcount decisions.

6.4 Top 10 Employees by Salary (Table)

This table lists the ten highest-paid employees along with their department and exact salary. The highest earner (Employee_210, Sales, \$112,790) and the runner-up (Employee_479, HR, \$101,183) are immediately visible. This visualization helps HR spot unusually high compensation outliers and can support conversations around pay equity or individual performance review.

6.5 Employee Distribution by Department (Pie Chart)

A pie chart shows the proportional share of each department within the organization. Marketing leads with 211 employees (21.1%), followed by HR with 205 (20.5%), Sales with 200 (20%), Finance with 193 (19.3%), and IT with 191 (19.1%). The distribution is notably balanced, with no single department dominating the workforce.

6.6 Employee Distribution by Gender (Donut Chart)

A donut chart visualizes the gender split: 50.9% female and 49.1% male. The near-equal split indicates that the organization has achieved reasonable gender balance at the aggregate level. The donut format makes the proportion immediately clear without requiring the viewer to read exact numbers first.

6.7 Employee Distribution by Age Group (Column Chart)

This column chart breaks the workforce into three age groups. The Older group (45+) has the most employees (approximately 462), followed by Middle Age (30–44, approximately 344), and Young (18–29, approximately 194). This distribution suggests a relatively senior workforce overall, which has implications for succession planning and knowledge transfer as older employees approach retirement age.

6.8 Attrition Rate by SalaryLevel (Column Chart)

This chart displays the attrition rate for each salary level. The Low salary group has the highest rate at 51.70%, followed by Medium at 50.43%, and High at 48.39%. Although the differences are not dramatic, the trend is consistent: employees in lower salary brackets leave at a slightly higher rate than those in higher ones.

6.9 Attrition Quantity by SalaryLevel (Column Chart)

While the previous chart shows rates, this one shows raw attrition counts. The Medium salary group has by far the most departures (352 employees), compared to Low (76) and High (75). This reflects the fact that the Medium group contains the majority of employees, so even with a slightly lower rate, the absolute number of departures is much larger.

6.10 Attrition Quantity by Gender (Column Chart)

This chart compares the number of attrited employees by gender: 259 female and 244 male. The difference is relatively small and may be partly explained by the fact that females slightly outnumber males in the workforce overall. No strong gender-based attrition pattern is apparent from this data alone.

6.11 Attrition Rate by Department (Column Chart)

Finance has the highest attrition rate at 54.00%, followed by HR at 52.68%, Sales at 50.78%, IT at 48.17%, and Marketing at 45.97%. Finance's elevated attrition stands out and may warrant further investigation into department-specific working conditions, workload, or compensation satisfaction.

6.12 Salary vs Performance Score (Scatter Plot)

This scatter plot places each employee as a point on a grid where the x-axis represents salary and the y-axis represents performance score. A trend line is overlaid on the data. The trend line is nearly horizontal, indicating a very weak positive relationship between salary and performance. Employees across all salary levels are distributed fairly evenly across performance scores 1 through 5, suggesting that higher pay does not reliably correspond to higher performance in this organization.

6.13 Average Salary by Department (Column Chart)

This chart shows the mean salary for each department. HR has the highest average at approximately \$57K, followed by Sales (\$56K), IT (\$55K), Marketing (\$55K), and Finance (\$53K). The gap between the highest and lowest is roughly \$4K, which is relatively modest but still notable given that Finance also has the highest attrition rate.

6.14 Experience Distribution by Age Group (Boxplot)

The boxplot shows the spread of ExperienceYears values within each age group. All three groups show a similar median of around 11–12 years, with broadly comparable interquartile ranges. The Young group naturally has a lower ceiling on experience than the Older group, though the similarity across groups is partly a function of how ExperienceYears was calculated from HireDate rather than from career start date.

6.15 Attrition by Experience Years (Line Chart)

This dual-line chart plots attrition status (Yes/No) against ExperienceYears. Both lines fluctuate across experience levels, and no clear, sustained pattern of attrition clustering at any specific experience level is immediately apparent. This suggests that attrition may not be strongly driven by length of service alone.

6.16 Attrition Rate by Experience Years (Line Chart)

This single-line chart shows the attrition rate at each experience year value. The rate fluctuates considerably, with several peaks reaching above 55–60% at certain experience levels. The

variability makes it difficult to draw firm conclusions, though the chart does suggest that some experience-year bands have notably elevated attrition risk.

6.17 Average Salary by Job Title (Bar Chart)

This horizontal bar chart displays average salary for a range of job titles, sorted in descending order. The chart provides HR with a quick reference for understanding how compensation varies across roles, which is useful for benchmarking and ensuring internal pay equity.

7. Analytical Findings

This section addresses the ten core business questions that guided the dashboard design. Each finding is derived directly from the visualizations and measures described above.

7.1 Which Department Has the Highest Number of Employees?

Marketing is the largest department, with 211 employees representing 21.1% of the total workforce. The five departments are distributed relatively evenly, with the smallest (IT, 191 employees) accounting for 19.1%. This suggests a fairly flat organizational structure without a dominant central function.

7.2 Which Department Has the Highest Attrition Rate?

Finance has the highest attrition rate in the organization at 54.00%. This means that over half of all Finance employees in the dataset have left the company. Notably, Finance also has the lowest average salary (\$53K) of all departments, which may be a contributing factor — though no causal relationship can be established from the available data alone.

7.3 Which Gender Is More Represented in the Company?

Female employees represent 50.9% of the workforce, while male employees account for 49.1%. The difference is small — approximately 1.8 percentage points — and the organization can be described as having a broadly equal gender balance at the aggregate level. Whether this balance holds within specific departments or job titles is a natural follow-up question that the slicers can help explore.

7.4 Which Department Has the Highest Average Salary?

HR has the highest average salary at approximately \$57,000 per year. This is closely followed by Sales at roughly \$56,000. Given that HR also has the second-highest attrition rate (52.68%), the relationship between salary and retention appears to be influenced by factors beyond compensation alone.

7.5 Which Department Has the Lowest Average Salary?

Finance has the lowest average salary at approximately \$53,000 per year. Combined with the highest attrition rate, this pattern may suggest that Finance employees feel less satisfied with their compensation relative to peers in other departments — though without exit interview data or employee satisfaction scores, this interpretation remains speculative.

7.6 Is There a Relationship Between PerformanceScore and Salary?

The scatter plot and overlaid trend line suggest a weak positive relationship at best. The trend line is nearly horizontal, and salary values are widely dispersed across all performance scores. Employees scoring a 1 on performance can earn as much as employees scoring a 5. This finding could indicate that salary increases are driven more by seniority, role, or negotiation than by individual performance outcomes. It may also signal an opportunity for HR to review compensation frameworks.

7.7 Which Age Group Is the Most Common?

The Older age group (45 and above) is the most common, with approximately 462 employees, representing nearly half the workforce. The Middle Age group (30–44) has around 344 employees, and the Young group (18–29) the fewest at approximately 194. This workforce profile suggests that a significant portion of employees may be approaching retirement in the coming years, which raises questions about succession planning and knowledge transfer.

7.8 Who Leaves the Company More Often — Younger or More Experienced Employees?

The Attrition by Experience Years and Attrition Rate by Experience Years charts do not show a clear, linear trend. Attrition appears across all experience levels without a strong concentration in any particular bracket. The data does not provide a definitive answer to whether younger or more experienced employees are more likely to leave. Based on the available data, attrition appears broadly distributed, which may indicate that the causes of departure are varied and not driven by a single demographic profile.

7.9 How Has the Hiring Trend Changed Over the Years?

The Employee Hiring by Year line chart shows fluctuation in annual hiring volumes from 2005 to approximately 2024. Hiring generally ranged between 30 and 60 employees per year, with noticeable dips around 2012–2013 and occasional peaks in other years. No clear sustained growth or decline trend is visible over the full period, which may reflect a relatively stable organizational size with cyclical recruitment needs. Without additional context, it is not possible to link specific hiring spikes or drops to organizational events.

7.10 In Which SalaryLevel Group Is Attrition Most Common?

This depends on how 'most common' is interpreted. By raw count, the Medium salary group has the highest attrition quantity with 352 departures — which reflects the fact that most employees fall within this salary band. By attrition rate, the Low salary group has the highest rate at 51.70%, compared to 50.43% for Medium and 48.39% for High. HR should be aware of both dimensions: while the Low group has the highest proportional risk, addressing attrition in the Medium group would have the greatest absolute impact on total employee numbers.

8. Key Insights

Drawing together the analytical findings above, the following insights emerge as the most business-relevant takeaways from this project. These are presented carefully, using conditional language where appropriate, as the dataset does not include all possible explanatory variables.

- **Attrition is exceptionally high overall.** An attrition rate of 50.30% means that one in every two employees in this dataset has left the company. This is a striking figure and may indicate structural workforce challenges, although without context on industry norms or time frame, it is difficult to assess how alarming this truly is.
- **Finance may need urgent attention.** Finance combines the lowest average salary with the highest attrition rate. While these two facts do not necessarily have a causal relationship, the pattern is worth investigating. It is possible that Finance employees feel undercompensated relative to market rates or relative to peers in other departments.
- **The workforce skews older.** With 46.2% of employees aged 45 or above, the organization faces a potential succession challenge in the medium term. If a significant portion of this group leaves within the next five to ten years, the organization may struggle to replace accumulated expertise and institutional knowledge.
- **Salary and performance appear weakly connected.** Based on the scatter plot, higher performance scores do not reliably correspond to higher salaries. This could suggest that the current compensation model is not strongly performance-driven, which may be a point of dissatisfaction for high performers who feel their contributions are not reflected in their pay.
- **Gender balance is strong at the aggregate level.** A near 50-50 gender split suggests that the organization has achieved broad gender parity in headcount. However, this aggregate figure does not reveal whether specific departments, salary levels, or job titles show gender imbalances — an analysis that slicers on the dashboard can facilitate.
- **Medium salary employees drive attrition in absolute terms.** Because the majority of employees fall within the Medium salary band, even a slightly lower attrition rate compared to the Low group still translates into far more departures in absolute numbers (352 vs 76). Retention strategies targeting the Medium salary group may therefore have the largest organizational impact.

9. Limitations

While the dashboard provides meaningful insight into workforce patterns, several limitations should be acknowledged when interpreting the findings.

- **No reason for leaving.** The dataset records whether an employee left (Attrition = Yes), but it does not capture why they left. Without exit interview data, it is impossible to determine whether attrition was voluntary or involuntary, and what the primary drivers of departure were.
- **No employee satisfaction data.** Measures such as job satisfaction scores, engagement survey results, or Net Promoter Scores would significantly enrich the analysis. Without them, explanations for patterns like Finance's high attrition remain speculative.
- **ExperienceYears is based on hire date only.** The ExperienceYears column reflects time since joining the current organization, not total work experience. Two employees with the same hire date are treated as having equal experience even if one entered with 15 years of prior experience and the other had none.
- **No promotion or career history.** The dataset contains only a single snapshot of each employee's job title and salary. Without promotion history or salary growth data, it is not possible to assess whether career progression varies by department or demographic group.
- **No workload or management data.** Factors such as average hours worked, manager quality, team size, or work-life balance are not captured. These are commonly cited drivers of attrition and their absence limits the explanatory power of the analysis.
- **Cross-sectional snapshot.** The dataset represents a single point in time (or a cumulative record of past employees). Without longitudinal tracking, it is not possible to analyze how individual employees' salaries, performance scores, or career trajectories evolved over time.

10. Recommendations

Based on the findings and the limitations identified above, the following recommendations are offered to guide HR's next steps.

10.1 Investigate Finance Attrition

Finance stands out with the highest attrition rate and the lowest average salary. HR should consider conducting structured exit interviews or stay interviews with current Finance employees to understand whether compensation, workload, management, or career development concerns are driving departures. If market benchmarking reveals that Finance salaries are below industry averages, a targeted salary review may be warranted.

10.2 Address Low Salary Group Attrition Rate

Although the Low salary group is smaller in absolute size, its attrition rate of 51.70% is the highest of any salary tier. HR should evaluate whether employees in this band have a clear path to salary progression and whether they feel valued within the organization. Small improvements in base pay or benefits for this group could yield meaningful retention gains.

10.3 Plan for Workforce Aging and Succession

With nearly half the workforce aged 45 or above, the organization should begin or accelerate succession planning efforts. Identifying critical roles held by Older employees and developing younger employees or external candidates to fill those roles over time will reduce the risk of knowledge loss. Mentorship programs pairing Older and Young employees could be particularly effective.

10.4 Review the Salary-Performance Link

The weak relationship between PerformanceScore and Salary may be worth examining more closely. HR could analyze whether high-performing employees (PerformanceScore 4–5) are more likely to leave than their lower-performing peers, and whether introducing more explicit performance-linked pay would reduce attrition among top contributors. At a minimum, the findings suggest that the current compensation model may not be perceived as merit-based.

10.5 Collect Richer Data for Future Analysis

To significantly improve the quality of future HR analytics, the organization should consider collecting: (a) reasons for leaving via structured exit interviews, (b) employee satisfaction or engagement scores collected annually, (c) promotion history and salary progression data, and (d) manager-level aggregates to identify teams with elevated attrition. With these additional

data sources integrated into Power BI, the dashboard could evolve from descriptive reporting into predictive workforce planning.

11. Conclusion

This project successfully delivered a fully interactive HR Analytics Dashboard in Power BI, built on a dataset of 1,000 employee records. Through structured data preparation in Power Query, the creation of meaningful calculated columns, and the development of seven core DAX measures, the raw data was transformed into a comprehensive analytical tool.

The dashboard addresses ten practical HR business questions and provides decision-makers with the ability to filter and explore workforce data across five key dimensions: department, gender, age group, performance score, and salary level. The visualizations span employee distribution, salary analysis, attrition patterns, hiring trends, and performance-salary relationships — offering a 360-degree view of the organization's human capital.

Among the most notable findings: the overall attrition rate of 50.30% is strikingly high; Finance emerges as a department of concern due to its combination of the lowest average salary and highest attrition rate; and the weak relationship between PerformanceScore and Salary raises questions about whether the current compensation model rewards performance effectively.

While the dataset has limitations — most notably the absence of reasons for leaving, satisfaction data, and career history — the dashboard provides a solid foundation for evidence-based HR conversations. It demonstrates how Power BI can turn routine HR data into actionable insights, enabling the organization to move from reactive decision-making toward proactive workforce management.

Ultimately, the value of any analytics dashboard lies not in the charts themselves, but in the questions they prompt and the decisions they inform. This HR Analytics Dashboard was built with that principle in mind — and it is hoped that it will serve as a useful starting point for a more data-driven approach to managing the organization's most important asset: its people.

— End of Report —